KØBENHAVNS UNIVERSITET

Computability and Complexity: Problem Set 4

Due: Thursday April 9 at 23:59 AoE.
Submission: Please submit your solutions via *Absalon* as a PDF file. State your name and e-mail address at the top of the first page. Solutions should be written in LaTeX or some other math-aware typesetting system with reasonable margins on all sides (at least 2.5 cm). Please try to be precise and to the point in your solutions and refrain from vague statements. Never just state an an answer, but make sure to also explain your reasoning. *Write so that a fellow student of yours can read, understand, and verify your solutions.* In addition to what is said below, the general rules for problem sets stated on the course webpage always apply.
Collaboration: Discussions of ideas in groups of two to three people are allowed—and indeed, encouraged—but you should always write up your solutions completely on your own, from start to finish, and you should understand all aspects of them fully. It is not allowed to compose draft solutions together and then continue editing individually, or to share any text, formulas, or pseudocode. Also, no such material may be downloaded from or generated via the internet to be used in draft or final solutions. Submitted solutions will be checked for plagiarism. You should also clearly acknowledge any collaboration. State close to the top of the first page of your problem set solutions if you have been collaborating with someone and if so with whom. *Note that collaboration is on a per problem set basis, so you should not discuss different problems on the same problem set with different people.*
Reference material: Some of the problems are "classic" and hence it might be possible to find solutions on the internet, in textbooks or in research papers. It is not allowed to use such material in any way unless explicitly stated otherwise. Anything said during the lectures or in the lecture notes, or any material found in Arora–Barak, should be fair game, though, unless you are specifically asked to show something that we claimed without proof in class. All definitions should be as given in class or in Arora–Barak and cannot be substituted by versions from other sources. It is hard to pin down 100% watertight, formal rules on what all of this means—when in doubt, ask the main instructor.
Grading: A total score of 70 points will be enough for grade 02, 105 points for grade 4, 140 points for grade 7, 175 points for grade 10, and 220 points for grade 12 on this problem set. Please note that problems are not necessarily presented in order of difficulty. Unless otherwise stated, every subproblem can be solved independently of other subproblems. Any revised versions of the problem set with clarifications and/or corrections will be posted on the course webpage `jakobnordstrom.se/teaching/CoCo26/`.
Questions: Please do not hesitate to ask the instructors or TA if any problem statement is unclear, but please make sure to send private messages when using Absalon—sometimes specific enough questions could give away the solution to your fellow students, and we want all of you to benefit from working on, and learning from, the problems. Good luck!

1 (10 p) Prove that any (non-constant) monotone Boolean function can be computed by a monotone Boolean circuit.

2 (40 p) In the monotone circuit lower bound for clique presented in the lectures, we used the concept of *sunflowers*, which are collections of sets $X_1, \ldots, X_p$ such that there is a set X with the property that $X_i \cap X_j = X$ for all $1 \leq i < j \leq p$. This is quite a strict requirement, and so one can ask whether the definition could be relaxed a bit without breaking the lower bound proof (and perhaps even yielding a slightly better bound).

2a Let us say that a collection of sets $X_1, \ldots, X_p$ is a *sub-sunflower* if there is a set X with the property that $X_i \cap X_j \subseteq X$ for all $1 \leq i < j \leq p$. Would the lower bound we did in class still go through if we did the "plucking" with sub-sunflowers? Please explain how to adapt the argument or point out where it fails.

2b Say that $X_1, \ldots, X_p$ form a *super-sunflower* if there is a set X such that $X_i \cap X_j \supseteq X$ for all $1 \leq i < j \leq p$. Would the lower bound we did in class still go through if we used super-sunflowers? Please explain how to adapt the argument or point out where it fails.

Remark: You should not expect to have to write pages of detailed arguments in the cases where you believe the proof still works or can be adapted to work. Also, when it seems that the proof cannot be made to work you do not have to prove beyond all doubt that no way of formalizing an argument along similar lines can possibly work in any universe—it is enough to point out, briefly but concretely, what technical difficulties arise, and why they seem hard to circumvent.

3 (50 p) In class, we defined Max-Exact-3-Sat as the problem where we are given a CNF formula ϕ such that every clause has exactly three literals, each on distinct variables, and we would like to find an assignment that satisfies as many clauses as possible. We saw that this problem has an easy randomized algorithm—pick a random assignment—with a respectable approximation guarantee of $\frac{7}{8}$. We also claimed that this algorithm can be made deterministic while retaining the approximation guarantee. In this problem, we will prove this claim.

3a Prove that expected number of clauses satisfied by a random assignment is $\frac{7}{8}m$, where m is the number of clauses.

3b Consider the sets of truth value assignments $A_0 = \{(a_1, \ldots, a_n) \in \{0,1\}^n \mid a_1 = 0\}$ and $A_1 = \{(a_1, \ldots, a_n) \in \{0,1\}^n \mid a_1 = 1\}$. Prove that either A_0 or A_1 must contain an assignment which satisfies at least a $\frac{7}{8}$ fraction of all clauses.

3c Use the observation in Problem 3b to find deterministically an assignment that satisfies at least a fraction $\frac{7}{8}$ of all clauses.

4 (40 p) A reduction from one gap problem GAP-A_α to another gap problem GAP-B_β for $0 < \alpha, \beta < 1$ is said to be a *gap-preserving reduction* if it maps YES-instances of GAP-A_α to YES-instances of GAP-B_β and NO-instances of GAP-A_α to NO-instances of GAP-B_β The existence of a gap-preserving reduction from GAP-A_α to GAP-B_β implies that if it is NP-hard to approximate problem A to within a factor α, then it is NP-hard to approximate problem B to within a factor β. This notion can also be extended to GAP-$A_{1-\varepsilon,\alpha}$ as explained further below.

Show that for every $\alpha > 0$ there exist ε, β, and a gap-preserving reduction from GAP-3-SAT$_\alpha$ to GAP-2-SAT$_{1-\varepsilon,\beta}$. Use this to conclude that there exists a $\beta \in (0, 1)$ such that approximating MAX-2-SAT to within a factor β is NP-hard.

Note: The gap problems GAP-3-SAT$_\alpha$ and GAP-2-SAT$_{1-\varepsilon,\beta}$ consider how many clauses in a CNF formula ϕ can be satisfied simultaneously by any assignment and are defined as follows.

GAP-3-SAT$_\alpha$:

$$\begin{aligned} \text{YES} &= \{\phi \mid \phi \text{ is a satisfiable 3-CNF formula}\} \\ \text{NO} &= \{\phi \mid \text{At most an } \alpha\text{-fraction of the clauses in the 3-CNF formula } \phi \text{ can be satisfied}\} \end{aligned}$$

GAP-2-SAT$_{1-\varepsilon,\beta}$:

$$\begin{aligned} \text{YES} &= \{\phi \mid \text{An } (1-\varepsilon)\text{-fraction of the clauses in the 2-CNF formula } \phi \text{ can be satisfied}\} \\ \text{NO} &= \{\phi \mid \text{At most a } \beta\text{-fraction of the clauses in the 2-CNF formula } \phi \text{ can be satisfied}\} \end{aligned}$$

5 (60 p) In class, we mentioned that Håstad proved a strengthening of the PCP Theorem, which shows that every language in NP has a PCP with 3 queries and soundness error close to $\frac{1}{2}$.

In this problem, we will prove that if we would be able to strengthen this further to show that some NP-complete language L is in $\mathsf{PCP}_{1,s}[O(\log n), 2]$, i.e., has a protocol with only 2 queries and some soundness error $s \in (0, 1)$, then it holds that $\mathsf{NP} = \mathsf{P}$.

5a Prove that if there exists an NP-complete language L that is in $\mathsf{PCP}_{1,s}[O(\log n), 2]$ for some $s \in (0, 1)$, then $L \leq_p$ GAP-2-SAT$_{1,\beta}$, where GAP-2-SAT$_{1,\beta}$ is as defined in Problem 4 but with $\varepsilon = 0$.

Hint: For the reduction, try to build a CNF instance which runs over all possible choices of the verifier's random bits, and checks if the verifier accepts the proof based on its 2 queries.

5b Suppose $G = (V, E)$ is a directed graph. Recall that for any pair of vertices u and v we say that they are in the same *strongly connected component (SCC)* if there are directed paths from u to v and from v to u in the graph G (or if $u = v$). For any directed graph G, we can partition the vertices into strongly connected components. Moreover, if we *contract* each such strongly connected component into a single vertex, then the resulting graph must be a directed *acyclic* graph (DAG); we will refer to it as the *underlying* DAG of the strongly connected components.

It is well-known (and is covered in introductory algorithms courses at DIKU) that for any directed graph G one can find the strongly connected components of G as well as the underlying DAG in polynomial time. Use this to show that 2-SAT has a polynomial time algorithm. For a full score, you need to both describe the algorithm and prove that it is correct, i.e., that it declares a 2-SAT instance satisfiable if and only if it is truly satisfiable.

5c Prove that gap-2SAT$_{1,\beta}$ is in P.

6 (60 p) This problem is about proving that a matrix that is close to the identity matrix has high rank (following the lecture notes).

6a Prove that for every matrix M of rank r and even $k > 0$ it holds that

$$\mathrm{rank}(A) \leq \binom{r+k}{k} ,$$

where the entries of A are $A_{ij} = M_{ij}^k$.

6b Prove that the following holds for large enough n. Assume that M is an $n \times n$ matrix so that $M_{ii} \geq 2/3$ for all $i \in [n]$ and $|M_{ij}| \leq 2/3$ for all $i, j \in [n]$, $i \neq j$. Also assume that the rank of M is at most $(\log_2 n)/1000$. Show that there is an $n \times n$ matrix A of rank $\leq n/10$ such that for every $i, j \in [n]$ it holds that

$$|A_{ij} - I_{ij}| \leq \frac{1}{n} ,$$

where I is the $n \times n$ identity matrix.

7 (60 p) For an $n \times n$ real matrix M, define the norm $\|M\|_\nu$ by

$$\|M\|_\nu = \inf \left\{ \textstyle\sum_z |c_z| : M = \sum_z c_z M_z \right\} ,$$

where the matrices M_z are rank-1 matrices such that $\|M_z\|_\infty \leq 1$. Prove that for the $n \times n$ identity matrix I it holds that $\|I\|_\nu = 1$. You can solve the case $n = 2$ for partial credit.

8 (60 p) Denote by Δ_n the set of $x \in [0,1]^n$ such that $\sum_{i=1}^n x_i = 1$. Then *von Neumann's minimax theorem* says that for every $A \in \mathbb{R}^{n \times m}$ it holds that

$$\max_{x \in \Delta_n} \min_{y \in \Delta_m} x^\top A y \;=\; \min_{y \in \Delta_m} \max_{x \in \Delta_n} x^\top A y .$$

Following the lecture notes, recall that $D_\mu(f)$ denotes the distributional communication complexity f over a distribution μ, and that $R(f)$ denotes the (public-randomness) randomized communication complexity of f (both with error probability $\frac{1}{3}$).

Using the minimax theorem, prove that the equality

$$R(f) = \max_\mu D_\mu(f)$$

holds for every two-party function f.

9 (60 p) Let X, Y be finite sets of size $|X| = |Y| = n$. A rectangle R is a set of the form $A \times B$ where $A \subseteq X$ and $B \subseteq Y$. For a function $f : X \times Y \to \{\pm 1\}$, define

$$disc_R(f) = \frac{1}{|X| \cdot |Y|} \left| \sum_{x \in A, y \in B} f(x,y) \right|$$

and

$$disc(f) = \max_R disc_R(f) ,$$

where the maximum is over all rectangles R. Prove that for every function f as above it holds that $disc(f) \geq \frac{1}{10\sqrt{n}}$.